Thesis Title

Thesis Subtitle



Joab Gray Kloppers

Department of Electrical Engineering
University of Cape Town
Rondebosch, Cape Town
South Africa

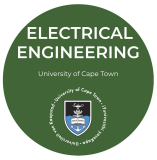
Supervisor
RA **Verrinder**

Co-supervisor(s)
A/Prof D **Pillay**, G **Maniel**

October 2026

BSc (Eng) final year project report submitted in partial fulfilment of the requirements for the degree of Bachelor of Science in Electrical and Computer Engineering in the Department of Electrical Engineering at the University of Cape Town

Keywords: keyword1; keyword2; keyword3



Declaration

I, **Joab Gray Kloppers**, hereby:

1. know that plagiarism is wrong. Plagiarism is to use another's work and pretend that it is one's own.
2. have used the **Harvard** convention for citation and referencing. Each contribution to, and quotation in, this report from the work(s) of other people has been attributed, and has been cited and referenced. Any section taken from an internet source has been referenced to that source.
3. declare that:
    a. this report is my own work, and apart from the normal guidance from my supervisor(s), I have received no assistance except as acknowledged.
    b. neither the substance nor any part of the above report has been submitted in the past, or is being, or is to be submitted for a degree at this University or at any other university.
    c. I have not paid a third party to complete my work on my behalf.
    d. my use of artificial intelligence software has been limited to:

    (specify which tool was used and how you used AI to assist with this project).

4. have not allowed and will not allow anyone to copy my work with the intention of passing it off as their own work.
5. acknowledge that copying someone else's work, or part of it, is wrong, and declare that this is my own work.
6. grant the University of Cape Town free license to reproduce the above thesis in whole or in part, for the purpose of research.

Word count [run latexmk] words (excluding preamble, appendices and bibliography)
Page count 23 pages (excluding preamble, appendices and bibliography)



Joab Gray Kloppers
Department of Electrical Engineering
University of Cape Town

Fri 9 October, 2026

Abstract

Thesis Title

Joab Gray **Kloppers**

Fri 9 October, 2026

Acknowledgements

Contents

Declaration i
Abstract ii
Acknowledgements iii
List of Figures vi
List of Tables vii
List of Symbols viii
List of Symbols viii
List of Acronyms and Glossary ix
Chapter 1 Introduction 1
1.1 Background 1
1.1.1 Biological Activity 1
1.1.2 Technical application 1
1.2 Problem statement 2
1.3 Project objectives 2
1.4 Scope, limitations and assumptions 2
1.5 Plan of development 2
Chapter 2 Literature review 3
2.1 Sandprawn Bioturbation and Ecological Significance 3
2.2 Surface Microtopography as a Proxy for Bioturbation 4
2.3 RGB-D Sensing for Non-Contact Surface Measurement 4
2.4 Summary and identified Research Gap 4
Chapter 3 Theoretical Backbone 6
3.1 Stereo Vision 6
3.2 Capture Alignment 6
3.3 Fusion 6
3.4 Evaluation 6
3.5 Roughness Calculation 6
3.6 Feature Detection 6
Chapter 4 System Design 7
4.1 Requirements Definition 7
4.2 System Overview 8
4.3 Hardware Rig 8
4.3.1 Frame 8
4.3.2 Camera, Tilt and Rotation Mechanism 8
4.3.3 AprilTag Boundary Markers 8
4.4 Software Architecture and Processing Pipeline 9
4.4.1 Capture Software 9

4.4.2 Cropping and Multi-Angle Fusion . . . 9
4.4.3 Microtopography Processing . . . 9
4.5 Evaluation Framework and Ground Truth . . . 10
4.5.1 Reference Surfaces . . . 10
4.5.2 Alignment and Metrics . . . 10
4.6 Summary . . . 10
Chapter 5 Intel RealSense Characterisation . . . 11
5.1 Plane Surface Characterisation . . . 11
5.1.1 Capture procedure and metrics . . . 11
5.1.2 Planar Characterisation Methodology . . . 12
5.1.3 Planar testing Results . . . 13
5.1.4 Discusion . . . 13
5.2 3D Reference Characterisation . . . 14
5.2.1 Reference objects and evaluation . . . 15
5.2.2 Testing Procedure . . . 15
5.2.3 3D Characterisation Results . . . 16
5.3 Chapter Summary . . . 16
Chapter 6 Data Processing Pipeline . . . 17
6.1 Depth Image Cropping . . . 17
6.2 Depth Image Fusion . . . 17
6.3 Surface Rugosity Estimation . . . 17
6.4 Feature Detection . . . 17
Chapter 7 Experimental Validation . . . 18
7.1 Known Relief and Controlled Targets . . . 18
7.2 Roughness Metric Validation . . . 18
Chapter 8 Results . . . 19
8.1 Characterisation Results . . . 19
8.2 Roughness Metric Results . . . 19
8.3 Feature Segmentation Results . . . 19
Chapter 9 Discussion . . . 20
9.1 Optimal Configuration . . . 20
9.2 Roughness Estimation . . . 20
9.3 Feature Detection . . . 20
Chapter 10 Conclusions . . . 21
Chapter 11 Further work . . . 22
11.1 Characterisation . . . 22
11.2 Hardware Choices . . . 22
Chapter A GitHub repository . . . 23
A.1 Title for appendix A.1 . . . 23
Appendix B Reference Objects . . . 24
Appendix B: Ethics approval . . . 26
Appendix C: Topic description . . . 27
Appendix D: ECSA Graduate Attributes . . . 28
Project mapping to ECSA Graduate Attributes . . . 28
Bibliography . . . 29

List of Figures

1.1 A simplified block diagram of a general communication system. 1

2.1 Custom incubation and flow through chamber by Schenone et al [1] 4

4.1 System overview. [TODO: Replace with a fuller diagram showing the data formats (.bag, .ply) passed between stages.] 8
4.2 Multi-angle fusion pipeline. 9

B.1 Large Egg Box Surface Reference Object. 24
B.2 Small Egg Box Surface Reference Object, Printed with amplitude A = 15mm. 24
B.3 Large Pyramid Surface Reference Object. 25

List of Tables

1.1 Table title 2

4.1 System requirements and where they are addressed. 7

4.2 Hardware summary. 9

5.1 Planar characterisation against a white wall at different working distances (one trial per distance; values averaged over the valid frames). 13

5.2 Planar characterisation under different lighting conditions at a height of approximately 0.9 m. Lux values are indicative (smartphone sensor). A dash indicates that no valid plane fit was obtained. 14

List of Symbols

Symbol	Description	Unit
$\boldsymbol{B}$	Magnetic flux density	T
$\boldsymbol{E}$	Electric field	V/m
$\boldsymbol{J}$	Current density	A/m^2
t	Time	s
ε_0	Permittivity of free space (8.85×10^{-12})	$C^2/N{\cdot}m^2$
μ_0	Permeability of free space ($4\pi \times 10^{-7}$)	N/A^2

Chapter 1

Introduction

1.1 Background

1.1.1 Biological Activity

Sand prawn activity has a large effect on the surrounding ecosystem. Pillay and Branch [2] explain that by physically burrowing into the sediment, they distribute buried sediment to the surface, alter the surface structure via the introduced mounds, which can increase surface area, and alter both chemical and structural properties of the sediment through burrowing and nutrient mixing. Venter et al [3] further shows that sand prawn bi-directional water pumping filters can reduce phytoplankton biomass by up to 70% in estuarine environments which can help with ecosystem resilience.

1.1.2 Technical application

Schenone et al [1] demonstrates that surface microtopography can be used as a reliable metric to predict biochemical processes. However common metrics for measuring surface roughness characteristics such as (list them here) are often invasive and result in the damage of the very structures under measurement. Azhar et al [4] demonstrated that the use of low cost RGB-D cameras can be used to produce depth images of the sediment surface, which can then be processed to analyse the surface properties.

Sub-subsection

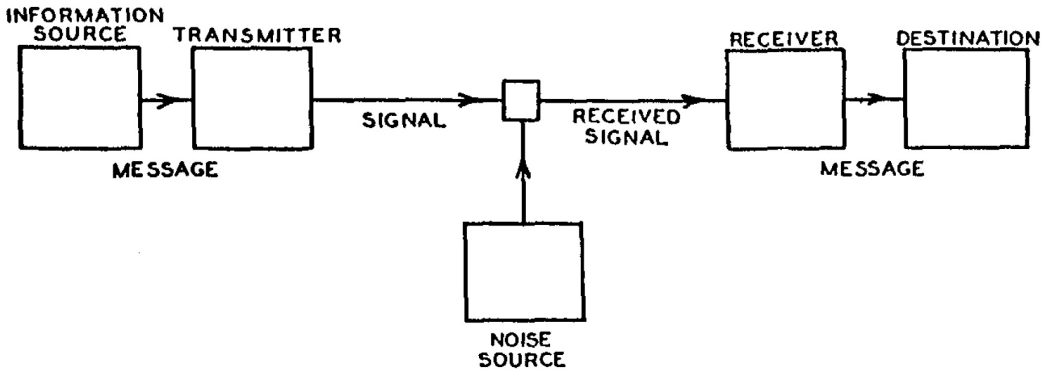


Figure 1.1: A simplified block diagram of a general communication system. A communication system can be abstracted as follows: an *information source* produces a message to be sent through a *transmitter* which converts it to a signal which can be sent via a communication *channel*. The received signal is noisy and is converted back to the message by the *receiver* which is then delivered to the *destination*. Taken from shannon1948

Table 1.1: Table title. Extended description of tables contents.

Heading 1	**Heading 2**	
	Subheading 1 [units]	*Subheading 2* [units]

1.2 Problem statement

Equations are numbered and include the chapter number. As they are considered part of the text, they include punctuation, and each parameter must be described in the text below. Ensure that the units are stated and where possible use SI units. For example, the Maxwell equations are expressed as:

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\epsilon_0}, \tag{1.1}$$

$$\nabla \cdot \boldsymbol{B} = 0, \tag{1.2}$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}, \tag{1.3}$$

$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \mu_0 \epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}, \tag{1.4}$$

where $\boldsymbol{B}$ is the magnetic flux density in T, $\boldsymbol{J}$ is the total current density in A/m^2, $\boldsymbol{E}$ is the electric field in V/m, μ_0 is the permeability of free space ($4\pi \times 10^{-7}$ H/m), and ϵ_0 is the permittivity of free space (8.85 $\times 10^{-12}$ C^2/N·m^2).

1.3 Project objectives

The Global Positioning System (GPS) is used for positioning. The Marginal Ice Zone (MIZ) is of particular interest in the Southern Ocean (SO).

1.4 Scope, limitations and assumptions

1.5 Plan of development

Chapter 2

Literature review

2.1 Sandprawn Bioturbation and Ecological Significance

Pillay and Branch [2] describe how sand prawn species are prominent members in the benthic ecosystem. The sand prawns continuously change the sediment properties such as surface topography porosity and biochemistry, through actions such as burrowing and water pumping. These actions redistribute buried sediment to the surface which results in surface mounds which increase the effective surface area of the sediment by up to 15 times. Kristensen et al [5] further explains how thalassinid shrimp populations can turn up to 12kg of sediment per square meter per day. These mounds can increase the effective hydrodynamic resistance which can help to stagnate the water flow close to the sediment. While the paper mentions measurements specific to sand prawn species located in other regions, they provide an insight into approximate feature sizes that can be expected, for example *Callianassa subterranea* present in the North sea can produce mounds 5.4cm tall and 11.7cm wide, and *Pestarella candida*, located in the Mediterranean and Adriatic Seas, which can produce mounds 3cm tall and 15cm wide.

Venter et al [3] further explains how sand prawns, *Callianassa kraussi*, present in the South African coastlines and at the Langebaan Lagoon, function as prominent water filterers. Within the burrows they actively irrigate the sediment through bi-directional pumping of water to replenish oxygen levels within the burrows, and to remove waste products. This irrigation has a secondary effect where phytoplankton are filtered out of the water by the burrow walls. This filtering can help improve ecosystem resilience to harmful conditions such as nutrient pollution which promote phytoplankton blooms. In relation to this project, the papers methodology entails manual visual counting of burrow openings.

The next question with regards to biological activity, is weather surface roughness, specifically rugosity, is a valid measurement to gauge the underlying biological activity within the benthic ecosystem. To this, Schenone et al [1, 6] determined experimentally that analysing two dimensional surface rugosity, they were able to correlate surface roughness measurements with sub-surface activity. One paper [1] described the testing of a controlled laboratory environment to simulate the benthic ecosystem, which can be seen in Figure 2.1. They found that the simple two dimensional rugosity estimate managed to account for up to 61% of nutrient variability. The second paper[6] describes a more complex implementation, with the use of a single diver operated camera recording video of the sediment surface, and the use of visual based structure from motion (SFM) photogrammetry to obtain a three dimensional representation of the surface. similarly, the find that sediment microtopography explains up to 55% of species richness variance, which is of particular interest with regards to this project. These papers show that surface rugosity is a useful measurement when gauging biological activity, and they provide a useful insight into the computational pipeline which is used to convert from spatial meshes to elevation matrices, however there are considerations that need to be considered, these papers discuss fully submerged environments, thus the requirement of under water cameras and SFM, whilst this project focuses on tidally exposed surfaces, where fixed clear observations can be made with the use of a rig.

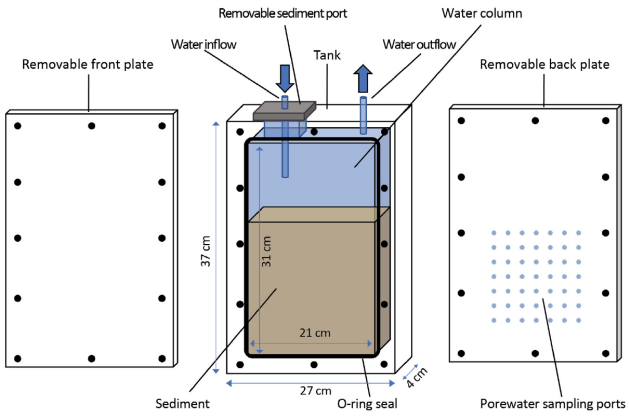


Figure 2.1: Custom incubation and flow through chamber by Schenone et al [1]

2.2 Surface Microtopography as a Proxy for Bioturbation

2.3 RGB-D Sensing for Non-Contact Surface Measurement

2.4 Summary and identified Research Gap

With the understanding that surface roughness statistics can provide a feasible proxy for benthic activity, there is a need for a method to scan sediment surfaces and extract topographical information. The current project, under the biological side **<PLEASE CHANGE THIS, SOMEHOW TIE IN THE EXISTING PROJECT>**, takes surface scans of static, exposed sediment, within the inter-tidal region in the Langebaan lagoon. These conditions mean that there is no need for submersion rated hardware, operated under dynamic conditions.

The current project is based off the work developed by Azhar et al [4]. The authors develop a framework that allows for extracting a millimetre scale 3D surface scans non-invasively, and at lower price scales than expensive LiDAR based systems. The system makes use of an off the shelf Asus Xiton pro Live 3D RGB-D structured light camera, mounted within an opaque plastic container to block infrared interference from sunlight. They then tested four polynomial de-trending algorithms to remove natural sediment gradients, and isolate the bioturbation. Due to the nature of the Asus Xiton, which makes use of structured light projection and a single IR camera, the system suffers when the surface is exposed to sunlight. Therefore the authors made use of an opaque plastic enclosure which is set on the sediment for scanning, however this limits the spatial resolution to only $0.3m^2$ with an effective camera resolution of only 240 x 200 pixels. Additionally the authors found that the higher order polynomial de-trending algorithms were sensitive to elevations at the edges of the image frame, requiring them to use a planar model which proved to be more robust.

Halmetschlager-Funek et al [7] and Rustler et al [8] provide useful insight by comparing several types of commonly available depth sensing cameras, encompassing different technologies such as structured light, active and passive stereo vision and time of flight (ToF). Halmetschlager-Funek et al [7] compares ten different sensors, with varying technologies and shows that sensor technologies like structured light and ToF generally struggle when exposed to high ambient light or reflective materials. These make it impractical to apply in the exposed environmental conditions considered for this project, as there is often direct sunlight, and small water pools which can increase reflectivity of the sediment. Rustler et al [8] compares four different stereo vision cameras, and finds that the Intel RealSense D435 is the superior camera for applications what are within one meter, however distance bias against planar surfaces increases significantly, compared to similar sensors such as the Intel RealSense D455, past 300cm distance. The author finds that for longer distance, the ZED 2 or Intel RealSense D455 are the best performers, with the main discerning factor being the image processing, as the ZED 2 requires dedicated GPU hardware for image processing, whereas the D455 does most of the processing on-board with some post precessing offloaded to the connected device.

Nguyen et al [9] discusses a method of deriving the lateral and axial noise profiles of the Microsoft Kinect using through analysis of specific scenes. To characterise lateral noise the authors make use of the fact the pixel statistical distributions along a depth edge are the same as statistical distribution of the point spreading factor (PSF), and thus can obtain a characterisation

on how points get blurred through the optics of the camera. For axial noise profiles, they analyse depth distributions across a plane of best fit, across various angles. Using these noise profiles, they are able to implement sensor specific filters that can filter noisy depth readings, whilst preserving geometric features. The benefit to this method is that it can be implemented across different sensors, enabling the application on the Intel RealSense D456 for improved depth filtering.

Metrological and critical characteriseation of intel d415 **<DONT THINK SHOULD USE THIS ONE>**

A benefit of using the Intel RealSense RGB-D cameras is the advanced ASIC which does all depth processing on device, thus eliminating the need for high performance computers during scans. Additionally the software development kit (SDK) for the RealSense family provides useful tools to the developer to tweak settings for specific applications, and to apply post-processing to scans for improved performance. Keselman et al, and Grunnet-Jepsen and Tong [10, 11] provide useful insight into the operation and theory of the Intel RealSense r200 and D400 series cameras respectfully. Keselman et al [10] describes how the correlation algorithm performs well under texture sparse conditions, and is more robust to noise compared to similar algorithms. The author also explains the limitations of the sensor, such as the minimum depth is dictated by the limited disparity search range, and maximum distance is limited by signal to noise ratio. Additionally, the use of a laser based feature projector can introduce some artifacting due to the effect of laser speckle noise. Grunnet-Jepsen and Tong [11] provide an outline on the post-processing methods provided by the Intel RealsSense SDK. This white paper explains how some of the processes work, and gives a general outline on the pipeline that should be applied to achieve a baseline filtered depth map.

One fundamental requirement of stereo vision is the need for textured surfaces, in order to perform disparity matching for depth computation. This need results as a reduction in performance on texture sparse environments. The Intel RealSense family address this with the use of a laser based IR projector which project semi random patterns onto the surface to artificially introduce visual texture. Grunnet-Jepsen et al [12] describes how the illumination power of the internal projector can be optimised for different light environments, and further how an external projector or dot pattern can be overlaid onto the surface to improve performance on texture sparse environments, as the D400 series cameras do not require a predefined pattern. These projected patterns can improve performance, as the achievable dot density is far higher than what the on board projector can achieve, and does not suffer with the same laser speckle problem.

<LOOK FOR A PAPER DISCUSSING TEXTURE OF WATER LOGGED SAND>

Intel Realsense camera depth testing methodology

pixel level extrinsics self calib??????

Chapter 3

Theoretical Backbone

3.1 Stereo Vision

3.2 Capture Alignment

3.3 Fusion

3.4 Evaluation

3.5 Roughness Calculation

3.6 Feature Detection

Table 4.1: System requirements and where they are addressed.

ID	Requirement	Addressed in
F1	Capture short-range depth scans of predominantly planar sediment	Sec. ??
F2	Restrict analysis to a repeatable, user-defined square region of the surface	Sec. ??, ??
F3	Capture the same surface from multiple viewpoints and fuse the scans into one reconstruction	Sec. ??
F4	Produce detrended microtopography, roughness metrics and segmented mounds and burrows	Sec. ??
E1	Allow working distance, viewing angle, lighting and depth-filter settings to be varied in a controlled, repeatable way	Sec. ??, ??
E2	Support characterisation of the camera's depth noise and error as a function of these conditions, to identify the optimal operating conditions for sediment	Chapter [TODO: ref]
E3	Measure the geometric error of fused scans under these conditions, and identify the optimal configuration for reconstruction	Sec. 4.5
C1	Use a single Intel RealSense D456	Sec. ??
C2	Rig is portable and adjustable in height for field deployment	Sec. ??
C3	Results are reproducible and comparable across runs and algorithms	Sec. 4.5

Chapter 4

System Design

This chapter defines the general requirements of the system as a whole and describes the hardware rig used to capture scans, and the software pipeline built to capture, and process the data to provide useful metrics about the sediment surface. The theory in Chapter [TODO: theory chapter ref] is applied here, and the sensor characterisation in Chapter [TODO: characterisation chapter ref] uses the system as specified below.

4.1 Requirements Definition

The aim of the overarching project is to quantify sandprawn bioturbation on intertidal sediment, within the Langebaan Lagoon, by measuring surface microtopography and extracting roughness metrics and visible biogenic features (mounds and burrow openings). The initial stages of the project were adapted from the framework of Azhar *et al.* [4], primarily modified to use the Intel RealSense D456.

This project aims to provide an investigation into the performance of the RealSense D456 camera currently used. The central aim is to characterise the depth camera in order to determine the optimal hardware and software configuration, and then to use scans taken under those configurations to build a high-quality reconstruction of the sediment on which the analysis is performed. The key output is an analysis of depth noise and reconstruction accuracy under different conditions.

Requirements are grouped as functional (F), experimental (E) and constraints (C), as shown in Table 4.1. Functional requirements state what the system as a whole must do. Experimental requirements state what system variables are investigated in the characterisation. Constraints are fixed conditions imposed on the design.



Figure 4.1: System overview. [TODO: Replace with a fuller diagram showing the data formats (.bag, .ply) passed between stages.]

4.2 System Overview

The system has three main operational stages, shown in Figure 4.1, with a fourth evaluation stage used for characterisation.

1. *Capture*: The capture software rs_marker scans the surface at multiple rotations and camera tilts.
2. *Fusion*: multi-angle scans are cropped to the tag-defined region, aligned and merged into a single reconstruction.
3. *Microtopography processing*: the fused surface is detrended, and roughness metrics and biogenic features are extracted.

For the purpose of characterisation, a fourth evaluation stage is used to gauge the reconstruction and metrics performance of the system against known target relief's.

4.3 Hardware Rig

4.3.1 Frame

The existing rig was built from 3030 aluminium extrusion in a $1\ \mathrm{m} \times 1\ \mathrm{m} \times 1\ \mathrm{m}$ frame. The top of the frame has a plastic tarp attatched, this allows for the shading of the scan area in the event of direct sunlight degrading the active stereo functionality of the D456. This frame was modified for the purpose of this project, in order to feature an H-shaped cross member that can slide vertically along the four legs. This enables the ability to adjust the camera height, and thus working distance, for purpose of characterisation and easy configuration. It was also modified to feature 3D printed April tag mounts that allow the position to be adjusted

4.3.2 Camera, Tilt and Rotation Mechanism

The Intel RealSense D456 is mounted on the H-frame through a 3D-printed bracket bolted to the extrusion. A servo motor in the bracket tilts the camera by up to $\pm 60°$. The camera clips onto a rotating platform on the tilt stage and can be turned manually through a full $360°$ in $45°$ increments. Together, tilt and rotation allow each surface point to be observed from several viewing directions, which reduces occlusion within burrows and mounds. This multi-view scanning additionally averages out view-dependent depth error. The servo is driven by an Arduino microcontroller, connected to and commanded from the operator's laptop, which also runs the capture software. The camera is operated at 1280×720 and 30 fps at a nominal working distance of [TODO: optimal distance].

4.3.3 AprilTag Boundary Markers

Four AprilTags are mounted on brackets on the frame legs, and each bracket can slide inward or outward. The tags' positions define a square region on the sediment surface to be processed and analysed. This gives a repeatable, physically marked analysis region, and its size can be changed to suit the desired scan region or height configuration of the camera. Because the tags are fixed to the frame, their detected positions also provide a geometric cue for aligning scans taken from different tilt and rotation settings.

Table 4.2: Hardware summary.

Component	Specification
Depth camera	Intel RealSense D456
Depth stream	1280×720, 30 fps
Working distance	≈700 mm
Frame	Aluminium extrusion, ≈1 m^3, sliding H-frame
Tilt actuation	Servo, ±60°, Arduino-controlled
Rotation	Manual, 45° increments, 360°
Boundary markers	Four adjustable AprilTags
Ambient light control	Overhead tarp

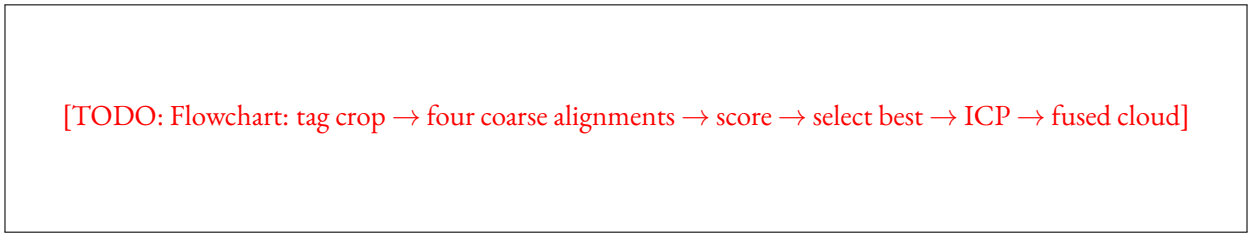


Figure 4.2: Multi-angle fusion pipeline.

4.4 Software Architecture and Processing Pipeline

4.4.1 Capture Software

Capture is implemented in C++ utilising the RealSense SDK. For each rig pose, it applies a configurable chain of depth post-processing filters and records the stream to a `.db3` file, together with IMU data. It detects the four AprilTags in the colour image, and the best frame of each pose is exported as a point cloud in PLY format. [TODO: State the filters used and their parameters, or reference the characterisation chapter.] Tag detection is stored along side captures for use of later processing through cropping and alignment.

4.4.2 Cropping and Multi-Angle Fusion

Each scan is first cropped to the square defined by the four tags. The cropped scans, taken at different tilt and rotation settings, are then fused (Figure 4.2). Fusion needs a good initial pose before refinement, so the pipeline generates several *coarse* alignment hypotheses by independent methods:

1. from the detected tag locations;
2. from the known rotation and servo tilt angle set on the rig;
3. from the camera IMU orientation;
4. from RANSAC matching of FPFH features [**?** **?**].

Each hypothesis is scored with a common alignment-quality measure, and the highest-scoring one is selected. It is then refined with ICP [**?**] and the merged surface passes to the microtopography stage. Using four independent seeds makes the fusion robust to the failure mode of any single method (for example, feature matching on a nearly planar surface, or IMU drift).

4.4.3 Microtopography Processing

The fused surface is detrended to remove the large-scale slope of the sediment, giving the microtopography from which roughness metrics are computed. Mounds and burrow openings are then detected and segmented using classical morphological image processing. [TODO: One sentence each on the detrending method, the roughness metrics, and the segmentation approach; details belong in later chapters.]

4.5 Evaluation Framework and Ground Truth

For characterisation and validation, a separate Python tool, `evaluate_scans`, measures the geometric error of any scan, whether a single capture or the fused output, against a known reference. It reuses the structure of the fusion pipeline's alignment stage, but aligns each scan to the reference. The tool then evaluates scan accuracy and coverage against the reference object.

4.5.1 Reference Surfaces

The reference objects used for validation include a 3D printed egg carton surface $z = sin(x) + sin(y)$, at large and small scale, and a square based pyramid. These reference objects provide a balance of condition matching of the actual sediment surface, using the soft sinusoidal surface, and general performance to less natural sharp shapes. For the purpose of validation, the 3D model files (.stl) are used for evaluation as the 3D mesh surface allows for point to surface distance calculation across the entire 3D object.

4.5.2 Alignment and Metrics

Alignment uses several seeded feature-matching starts, systematic perturbations of the best pose (quarter turns, a flip, and slides of a quarter and half of the part), and point-to-plane ICP, followed by a final refinement against the exact surface. Candidate poses are ranked by a truncated least-squares cost computed in both directions (scan to reference and reference to scan), which prevents a pose that matches fewer points from winning by default. This matters on periodic targets such as the egg carton, where several look-alike poses exist. The tool flags a result as *ambiguous* when the two best poses are too close to call, and reports degrees of freedom the surface cannot constrain.

For each aligned scan the tool reports accuracy (RMSE, MAE, bias, standard deviation, trimmed statistics, percentiles), the vertical error for height-field references, and coverage of the reference. It also reports completeness, precision and F-score at several distance thresholds, Chamfer and Hausdorff distances, outlier fraction and point density. Every run appends a row to a shared table with the exact parameters, reference hash and random seed, so experiments are reproducible and comparable across configurations.

4.6 Summary

The rig provides repeatable, multi-angle capture of a user-defined surface region. The software fuses those scans, extracts microtopography, and is checked against analytic ground truth. Chapter [TODO: characterisation chapter ref] uses this system to characterise the D456's depth noise.

Chapter 5

Intel RealSense Characterisation

The aim of the overarching project is to produce high-accuracy reconstructions of intertidal sediment from which surface roughness metrics can be extracted. The accuracy of those metrics is limited by the depth data they are computed from, so the Intel RealSense D456 was characterised before the capture and processing pipeline was finalised. The characterisation answers four practical questions about how the camera and rig should be operated in situ: at what working distance, at what viewing angle, under what ambient lighting, and with which depth-stream and filter settings the camera produces the most accurate surface.

The characterisation was performed in two stages. In the first stage the camera was tested against flat planar targets. A plane has an exact, easily measured ground truth, which allows the effect of working distance, tilt angle and ambient light to be isolated from the effect of surface shape. These tests give a baseline for sensor noise and provide the operating envelope used in the second stage. In the second stage the camera was tested against 3D-printed reference objects with known geometry (see Appendix B.) These tests identify the camera configuration, post-processing filter settings and combination of viewing directions that give the best reconstruction, using surface relief that is representative of the sediment of interest. Each characterisation below is presented with its method, results and a short discussion.

5.1 Plane Surface Characterisation

5.1.1 Capture procedure and metrics

All planar characterisation used custom C++ software built on the Intel RealSense SDK, following the plane-fit metrics of Intel's Depth Quality Tool. Details of each trial including: distance, tilt angle and a lux reading, are entered into the tool and the scan is started, after which approximately 600 frames are recorded. A region of interest (ROI) is then applied whose pixel size is adjusted with distance so that it always covers a fixed 50 × 50 cm area of the target. A plane is fitted to the ROI points of every frame and compared with the ground-truth plane. Frames flagged as repeated or as outliers were discarded, leaving roughly 400 to 600 valid frames per trial[TODO: confirm the same frame rejection was applied in all three sweeps]. Several metrics are reported by the tool, however there are four main metrics that will be discussed here to represent the general performance, each averaged over the valid frames of a trial:

- **Plane-fit RMS** (mm): the root-mean-square distance of the ROI points from the best-fit plane. This measures spatial noise and is independent of any constant offset in the depth.
- **Median depth error** (mm): the median signed difference between the measured and ground-truth depth, which measures systematic bias.
- **Fill rate** (%): the proportion of ROI pixels that returned a valid depth value.
- **Plane Distance Error** (mm): Error between perpendicular distance from camera to the fitted plane, and the ground truth.[TODO: is this a good metric?]

The distinction between precision and bias matters for this project. A smooth, systematic bias is removed when the fused surface is detrended to extract microtopography, whereas random noise adds directly to the measured roughness. Plane-fit

RMS and fill rate are therefore treated as the primary metric, with depth error and Plane Distance Error treated as secondary indicators. Each condition was recorded as a single trial. The frames within a trial capture temporal variation but not the repeatability of re-setting the experiment, which is discussed in section 5.3.

5.1.2 Planar Characterisation Methodology

Working distance

The camera was mounted on a tripod and pointed at a white, featureless wall, as recommended by Intel for depth-quality testing. This forces the camera to depend on its own projected infrared pattern. The tripod was used instead of the experimental rig because the rig's mounting brackets could not hold the camera sufficiently parallel to the floor: the camera angle varied with every scan and consistently exceeded 1.5°, which would have contaminated a distance sweep. The camera was positioned at eight distances between 604 mm and 903 mm, and the angle between the camera and the wall was minimised by hand at each position. The fitted-plane angle confirmed that the residual pitch and roll stayed below 0.5°in all trials. The ground-truth distance was measured between the wall and the cameras glass front using a tape measure.

[TODO: Maybe talk more about methodology?]

Camera tilt

The camera was mounted on the experimental rig so that its angle could be set with the tilt servo. A large flat white foam board was placed on the floor and re-centred in the field of view (FOV) after each change of angle. Two rotations were tested independently. The pitch axis is parallel to the line through the two IR imagers, and the roll axis is perpendicular to both the pitch axis and the optical axis. The angle was changed in steps of approximately 2°, covering pitch angles from $-27°$ to $-2°$ and roll angles from $+3°$ to $+25°$ for 26 trials in total[TODO: confirm that the opposite signs were not recorded, or add them]. The camera mount height was kept fixed. For each trial the vertical height of the left IR imager, the centre of the resulting depth image, above the board was measured, and the axial ground-truth distance was calculated from that height and the tilt angle. This axial distance varied between 657 mm and 743 mm over the sweep. The angle of the fitted plane agreed with the set angle to within 1° in every trial (mean 0.4°), which verifies the angle settings.

[TODO: Maybe talk more about methodology?]

Ambient lighting

The same foam board was scanned at a camera height of approximately 900 ,m under a range of lighting conditions, with the camera exposure left on automatic. Indoor conditions ranged from a dark room in the evening (blinds closed, lights off) to an afternoon room with the blinds open and the lights on. Outdoor trials were performed around midday in shade, in shade within direct sunlight, and in direct sunlight. The outdoor trials were taken twice per condition, one with the foam board, and one without the foam board. This was done as the effects of the high ambient light resulted in exposure settings that were unable to detect the cameras built in projected pattern, and thus the featureless white surface lost detection. The brick flooring was able to be detected in this condition, thus the results for the outdoor trials provide relational performance compare to the rest of the trials. Illuminance was read with a smartphone running the Physics Toolbox Suite application, which uses the front camera as a lux meter. Such readings are uncalibrated and are used here to indicate the order of magnitude and ranking of conditions, not precise values. The lux was recorded as 0 in the darkest condition and 24 400 in full sun.

[TODO: Maybe talk more about methodology?]

5.1.3 Planar testing Results

Working Distance

Results from this characterisation are shown in Table 5.1 [TODO: get statistical results for these measurements]. These are then plotted in [TODO: reference plots]

Table 5.1: Planar characterisation against a white wall at different working distances (one trial per distance; values averaged over the valid frames).

Distance (mm)	Plane-fit RMS (mm)	RMS (% of dist.)	Median depth error (mm)	Fill rate (%)	Plane dist. error (mm)
604	1.22	0.20	4.7	99.6	1
628	1.29	0.20	4.7	99.5	1
667	1.56	0.23	6.4	99.5	3
723	2.64	0.36	5.5	97.7	4
760	2.81	0.37	6.0	97.7	5
821	3.27	0.39	7.7	97.3	6
880	4.06	0.46	7.2	96.9	7
903	4.58	0.50	8.0	96.5	8

[TODO: Figure: plane-fit RMS and fill rate versus distance, with the power-law fit overlaid]

Camera Tilt

[TODO: Figure: plane-fit RMS versus tilt angle for pitch and roll, with the range-predicted curve overlaid] Plane-fit RMS was lowest near 0° (about 1.6 mm) and rose to 2.30 mm at $-27°$ pitch and 2.12 mm at $+25°$ roll. The fill rate stayed between 98.4% and 99.5%. The median depth error varied between 5 mm and 11 mm without a clear dependence on angle. Trials at almost the same angle differed by up to 6 mm (5.0 mm at $-26.8°$ against 10.7 mm at $-27.3°$ pitch).

Ambient Lighting

The results are listed in Table 5.2. Indoors, plane-fit RMS increased with illumination from 1.80 mm in darkness to 4.44 mm at 151 lux, and the fill rate fell from 99.9% to 96.3%. Adding artificial light increased the noise within both the evening session (1.80 to 2.84 mm) and the afternoon session (2.73 to 3.44 mm). Outdoors, the foam board in shade (388 lux) returned only 86.5% valid pixels and a plane-fit RMS of 6.00 mm. In direct sunlight (2 650 and 24 400 lux) no valid plane could be recovered, with fill rates of 3.7% and 5.4%. The ground surface without the board returned a 100% fill rate in all three outdoor conditions, including direct sun at around 10 000 lux, with a plane-fit RMS of 1.1 to 1.2 mm.

5.1.4 Discusion

Working Distance

Stereo triangulation noise is expected to grow with the square of the range. A power-law fit to these data gives an exponent of approximately 3.3, which is steeper than expected. Part of the difference may come from the ambient light, which was not constant across the sweep. The auto-exposure value dropped from around 5300 for the three closest distances to between 1700 and 2050 for the remainder, and the largest step in noise (from 1.56 mm to 2.64 mm) coincides with this change. The trials were recorded in a different order to their distance (between about 15:40 and 17:35), so distance and ambient light are partly confounded. Even allowing for this, the trend is clear: noise grows rapidly with working distance, so the camera should be operated at the shortest distance that still frames the analysis region. The roughly constant relative depth error of about 0.8% is consistent with a scale or ground-truth referencing offset rather than a distance-dependent effect. It does not affect the roughness metrics, but it would appear as a scale error in an absolute reconstruction, which the 3D reference tests in Section **??** quantify.

Table 5.2: Planar characterisation under different lighting conditions at a height of approximately 0.9 m. Lux values are indicative (smartphone sensor). A dash indicates that no valid plane fit was obtained.

Surface	Condition	Lux	Fill rate (%)	Plane-fit RMS (mm)	Median depth error (m
Indoor foam board	Evening, blinds closed, lights off	0	99.9	1.80	4.9
	Afternoon, blinds closed, lights off	17	99.4	2.73	7.3
	Evening, blinds closed, lights on	65	99.3	2.84	5.2
	Afternoon, blinds closed, lights on	78	98.7	3.44	7.3
	Afternoon, blinds open, lights off	91	97.0	4.14	7.1
	Afternoon, blinds open, lights on	151	96.3	4.44	7.5
Outdoor foam board	Shade	388	86.5	6.00	−2.5
	Direct sun, under shade	2 650	3.7	–	–
	Direct sun	24 400	5.4	–	–
Outdoor, no board	Shade	213	100	1.20	5.7
	Direct sun	9 800	100	1.09	4.6
	Direct sun, under shade	10 000	100	1.17	4.3

Camera Tilt

Because the camera pivots, tilting it also increases the axial distance to the board. When the power law from the distance sweep is applied to the measured axial distance of each trial, the predicted plane-fit RMS matches the observed value to within about 7% (for example 2.42 mm predicted against 2.30 mm observed at $-27.3°$, and 2.04 mm against 2.12 mm at $+25.3°$). The increase in noise with tilt over $\pm 25°$ can therefore be explained almost entirely by the longer range, and tilt itself does not appear to degrade the depth significantly. This is the result needed to justify multi-view scanning, since oblique views add information about occluded relief at little cost in noise. The scatter in depth error between neighbouring angles, which is larger than any trend, points to the repeatability of the experimental set-up (re-positioning the board and measuring its height) as the dominant source of uncertainty in this metric, and the depth-error values should be read with that in mind. A flat board cannot reveal the effects of foreshortening and self-occlusion on relief, which are examined with the 3D reference objects.

Ambient Lighting

Two conclusions follow. First, ambient light degrades the depth from active stereo even when the surface is well within the sensor's range, because the projected infrared pattern has to compete with the ambient light for contrast. Daylight appears more damaging per lux than artificial light (4.14 mm at 91 lux of daylight against 2.84 mm at 65 lux of artificial light), which is consistent with the greater near-infrared content of sunlight, so illuminance in lux is only a rough predictor of performance. Second, the surface matters as much as the illumination. The board failed at 2 650 lux, whereas the ground surface returned complete, low-noise data at 9 800 lux, so the failure cannot be attributed to the light level alone. A bright, featureless, highly reflective white surface is a worst case for stereo matching. The ground surface supplies natural texture, so the planar results on white targets should be read as pessimistic bounds on the noise expected on sediment. The outdoor tests did not include a matched pair of lux levels for the two surfaces, and wet or glossy sediment, which is more specular, was not tested. These results support retaining the shading tarp on the rig, which the Chapter **??** design already includes, to limit the ambient infrared reaching the scan area[TODO: check the label of the System Design chapter].

5.2 3D Reference Characterisation

The planar tests establish the sensor's noise behaviour but not how accurately it reconstructs three-dimensional relief, which is affected by resolution, filtering, foreshortening and the fusion of several views. These effects were measured against reference objects with known geometry using a modified version of the main capture program, which records scans under controlled software settings and hardware set-ups.

5.2.1 Reference objects and evaluation

Three 3D-printed reference objects were used (Appendix [TODO: ref]). Each has a 100 × 100 mm footprint. Two are sinusoidal "egg carton" surfaces of the form $z = \sin(x) + \sin(y)$ at a large scale (feature spacing 50 mm) and a small scale (25 mm), and the third is a square-based pyramid of 50 mm height[TODO: confirm the egg carton amplitude and wavelength against the CAD drawings]. The smooth sinusoidal surfaces approximate the mounds produced by sand prawns, whose heights of 3 to 5 cm were reported in the literature review, and the small carton tests the lateral resolution of the camera. The pyramid has sharp edges and planar facets, which show whether the filters preserve geometric edges or smooth them.

Scans were evaluated with the `evaluate_scans` tool described in Chapter **??**[TODO: Section ref]. The tool rigidly aligns each scan to the reference .stl model, using feature-matching starts followed by point-to-plane ICP, and then computes point-to-surface distances. Scale is not estimated, so any scale error in the camera data appears as geometric error. The metrics used are RMSE, mean absolute error, bias and standard deviation of the point-to-surface distance, the completeness, precision and F-score at distance thresholds of 0.5, 1, 2 and 5 mm, and the fraction of the reference surface covered. Every evaluation logs its parameters, reference hash and random seed, so all results can be reproduced and compared directly.

5.2.2 Testing Procedure

Camera configuration

The first sweep fixed the height and tilt of the camera and scanned the large egg carton and the pyramid with different stream configurations. The resolution ranged from 1280 × 720 to 640 × 360 pixels, the depth units from 0.001 m to 0.0001 m, and all of the built-in visual presets were tested except the Hand preset, which is intended for gesture tracking. The depth unit sets the quantisation step of the 16-bit depth values, and the finest setting (0.1 mm) still provides a maximum range of 6.5 m, well beyond the working distance. [TODO: Results table: RMSE, completeness and F-score for each resolution, depth unit and preset on both objects, plus one or two sentences of discussion]

Based on this sweep, a resolution of 1280 × 720, a depth unit of 0.0001 m and the High Density preset were used for all later scans[TODO: confirm that this was the outcome of the sweep]. These are the settings listed in Chapter **??**.

Post-processing filters

With the configuration fixed, five stationary scans of each object were recorded at each of the tested heights, without tilt or rotation. The data were recorded to file so that the RealSense post-processing filters could be applied offline in many combinations to the same scans (Grunnet-Jepsen and Tong [TODO: cite]). The filters were decimation, spatial, temporal and hole-filling, and a sweep was made over their smoothing alpha, smoothing delta and magnitude parameters[TODO: list the values or ranges tested for each filter]. Decimation reduces the resolution, the spatial filter performs edge-preserving smoothing, the temporal filter averages depth across frames, and hole filling interpolates invalid pixels. Each combination was evaluated against the reference, and the process was repeated at each height to determine whether the best filter settings depend on the working distance. [TODO: Results: best-performing settings per height and object, and the trade-off between noise reduction and loss of fine relief or edges]

Tilt and rotation

Finally, a full scan of each object was recorded at rotations of $-45°$, $0°$, $45°$ and $90°$ and at tilts from $-25°$ to $+25°$ in steps of $5°$. This gives 44 views of each object. Offline, these views were aligned and fused by the pipeline in Chapter **??**. For each view, several coarse alignment hypotheses are generated (from the IMU orientation, the detected AprilTag positions, the known rotation and tilt settings, and RANSAC matching of FPFH features) and scored. The best hypothesis is refined with ICP and the aligned scans are merged[TODO: expand once the fusion method is finalised]. Subsets of the views were fused and each fusion was evaluated against the reference, which shows how the reconstruction accuracy depends on the number of views and on the combination of tilt and rotation. [TODO: Results: accuracy and coverage versus the views included, and the recommended tilt and rotation set]

5.2.3 3D Characterisation Results

Camera Configuration

Filters

Tilt and Rotation

5.3 Chapter Summary

This chapter characterised the Intel RealSense D456 in two stages. The planar tests showed that depth noise grows rapidly with distance (from 1.2 mm at 604 mm to 4.6 mm at 903 mm), so the working distance should be as short as the analysis region allows. Tilts of up to about $\pm 25°$ add little noise beyond that caused by the increased range, which supports a multi-view strategy. Ambient light increases noise, and the white foam board failed in direct sun while a textured ground surface did not, so the tarp shading and the textured nature of sediment are both important. The 3D reference tests then fix the stream configuration, filter settings and view combinations used in the remainder of the project.

The results have limitations that affect how they should be used. Each condition was recorded once, so the effect of re-setting the experiment is not quantified, and the scatter in depth error between neighbouring tilt angles suggests that it is significant. The distance sweep was not recorded in order of distance and the lighting changed between sessions, so distance and ambient light are partly confounded. The lux values come from an uncalibrated smartphone sensor. The distance sweep was performed with a tripod because the rig brackets could not hold the camera level, and the white featureless planar targets are likely to overstate the noise expected on textured sediment. Wet and specular sediment has not yet been tested. These points are taken up in Chapter **??**[TODO: ref to Further work].

Chapter 6

Data Processing Pipeline

6.1 Depth Image Cropping

6.2 Depth Image Fusion

6.3 Surface Rugosity Estimation

6.4 Feature Detection

Chapter 7

Experimental Validation

7.1 Known Relief and Controlled Targets

7.2 Roughness Metric Validation

Chapter 8

Results

8.1 Characterisation Results

8.2 Roughness Metric Results

8.3 Feature Segmentation Results

Chapter 9

Discussion

9.1 Optimal Configuration

9.2 Roughness Estimation

9.3 Feature Detection

Chapter 10

Conclusions

Chapter 11

Further work

11.1 Characterisation

11.2 Hardware Choices

Appendix A

GitHub repository

This appendix provides a link to the GitHub repository where the code and additional materials for this final year project report can be found.

A.1 Title for appendix A.1

- GitHub Repository: Repo name

Appendix B

Reference Objects

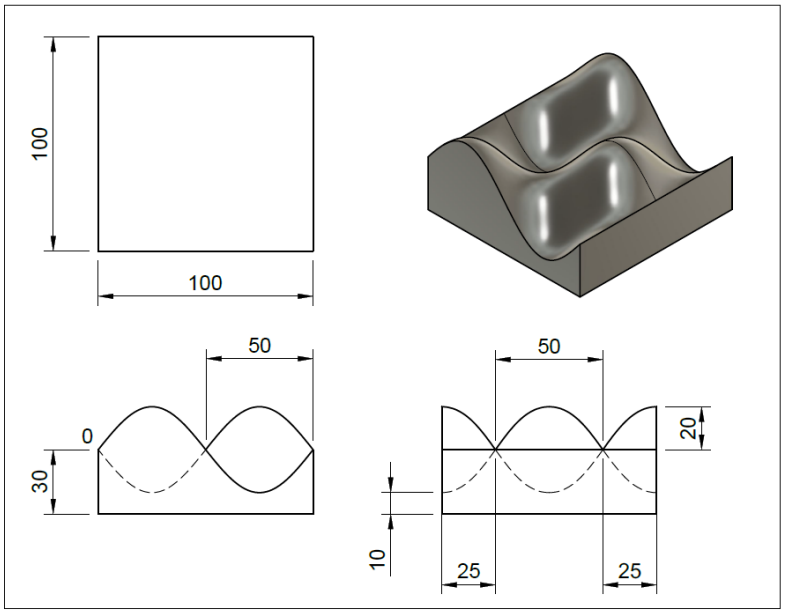


Figure B.1: Large Egg Box Surface Reference Object.

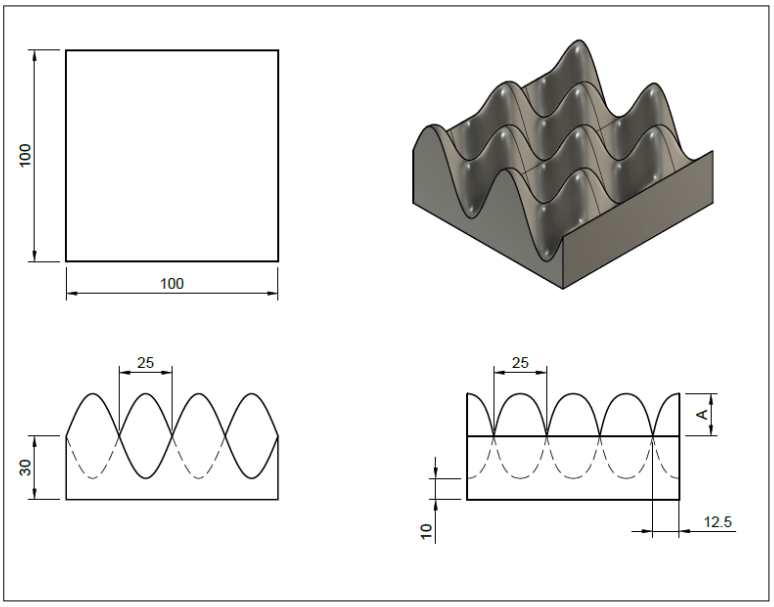


Figure B.2: Small Egg Box Surface Reference Object, Printed with amplitude A = 15mm.

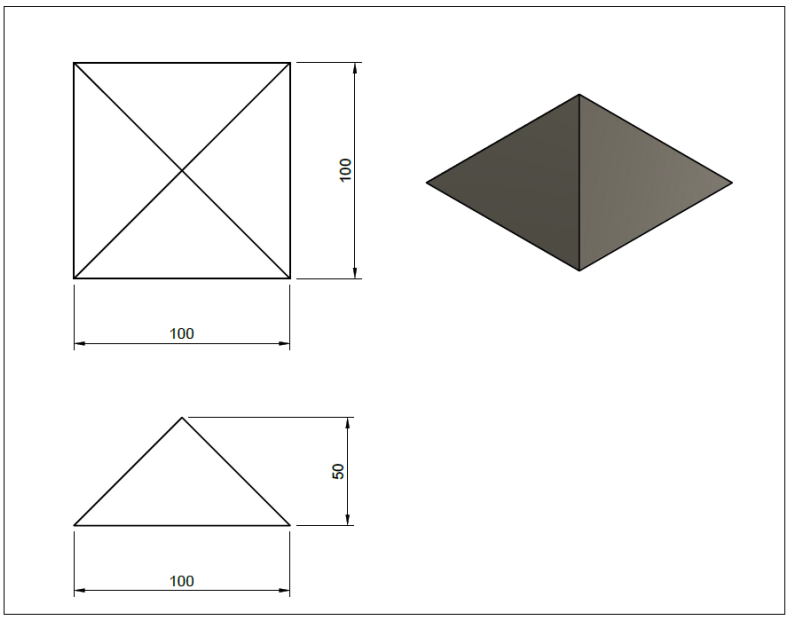


Figure B.3: Large Pyramid Surface Reference Object.

Appendix B: Ethics approval

Appendix C: Topic description

The original topic description used during the project selection process is included below. This topic has since been further refined, and the exact problem that was investigated is described in Section 1.2.

Appendix D: ECSA Graduate Attributes

Project mapping to ECSA Graduate Attributes

The table below provides a structured reflection on where each of the required Graduate Attributes (GAs) as defined by Engineering Council of South Africa (ECSA) associated with the final year project is met. This includes reference to the specific Chapter/Section (where applicable).

GA	Graduate Attribute Description	Student reflection: How this outcome was achieved during the project
GA1	*Problem solving*	
GA4	*Investigations, experiments and data analysis*	
GA5	*Use of engineering tools*	
GA6	*Professional and technical communication*	
GA8	*Individual and collaborative teamwork*	
GA9	*Independent learning ability*	

Bibliography

[1] S. Schenone, I. Bartl, and S. F. Thrush, "Sediment microtopography predicts localised benthic ecosystem functioning and the effect of species interactions," *Marine Ecology Progress Series*, vol. 749, pp. 47–55, Nov. 2024.

[2] D. Pillay and G. Branch, "Bioengineering effects of burrowing thalassinidean shrimps on marine soft-bottom ecosystems," in *Oceanography and Marine Biology: An Annual Review*, Jun. 2011, vol. 49, pp. 137–192.

[3] O. Venter, D. Pillay, and K. Prayag, "Water filtration by burrowing sandprawns provides novel insights on endobenthic engineering and solutions for eutrophication," *Scientific Reports*, vol. 10, no. 1, p. 1913, Feb. 2020.

[4] M. Azhar, J. R. Hillman, T. Gee, S. Schenone, W. van der Mark, S. F. Thrush, and P. Delmas, "An RGB-D framework for capturing soft-sediment microtopography," *Methods in Ecology and Evolution*, vol. 13, no. 8, pp. 1730–1745, 2022.

[5] E. Kristensen, G. Penha-Lopes, M. Delefosse, T. Valdemarsen, C. O. Quintana, and G. T. Banta, "What is bioturbation? The need for a precise definition for fauna in aquatic sciences," *Marine Ecology Progress Series*, vol. 446, pp. 285–302, Feb. 2012.

[6] S. Schenone, J. E. Hewitt, J. Hillman, R. Gladstone-Gallagher, J. Gammal, C. Pilditch, A. M. Lohrer, E. Ferretti, M. Azhar, P. Delmas, and S. F. Thrush, "Seafloor sediment microtopography as a surrogate for biodiversity and ecosystem functioning," *Ecological Applications*, vol. 35, no. 1, p. e3069, 2025.

[7] G. Halmetschlager-Funek, M. Suchi, M. Kampel, and M. Vincze, "An Empirical Evaluation of Ten Depth Cameras: Bias, Precision, Lateral Noise, Different Lighting Conditions and Materials, and Multiple Sensor Setups in Indoor Environments," *IEEE Robotics & Automation Magazine*, vol. 26, no. 1, pp. 67–77, Mar. 2019.

[8] L. Rustler, V. Volprecht, and M. Hoffmann, "Empirical Comparison of Four Stereoscopic Depth Sensing Cameras for Robotics Applications," *IEEE Access*, vol. 13, pp. 67 564–67 577, 2025.

[9] C. V. Nguyen, S. Izadi, and D. Lovell, "Modeling Kinect Sensor Noise for Improved 3D Reconstruction and Tracking," in *Visualization & Transmission 2012 Second International Conference on 3D Imaging, Modeling, Processing*, Oct. 2012, pp. 524–530.

[10] L. Keselman, J. I. Woodfill, A. Grunnet-Jepsen, and A. Bhowmik, "Intel(R) RealSense(TM) Stereoscopic Depth Cameras," in *2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, Jul. 2017, pp. 1267–1276.

[11] A. Grunnet-Jepsen and Tong, "Depth Post-Processing for Intel® RealSense™™ D400 Depth Cameras," https://www.intel.com/content/www/us/en/content-details/842031/depth-post-processing-for-intel-realsense-d400-depth-cameras.html.

[12] A. Grunner-Jepsen, J. N. Sweetser, P. Winer, A. Takagi, and J. Woodfill, "Projectors for D400 Series Depth Cameras."